

Latent-class and mixture-model formulations for biomarker-treatment interaction in N-of-1 trials

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1 Abstract

Draft placeholder. Architectures A and B in the companion paper encode the biomarker-treatment interaction as either a deterministic moderation of the mean (A) or a covariance-structure correlation that erodes under carryover (B). A reviewer of that paper raised a substantive question: if Architecture B is functionally an imperfect responder indicator, why is the heterogeneity not modelled explicitly, at the participant level, via a latent-class or mixture-model construction? This paper takes that suggestion as its starting point. We develop the latent-class and finite-mixture formulations of the biomarker-treatment interaction problem, characterise their identifiability and estimability under the N-of-1 family of trial designs, and contrast their statistical properties against Architectures A and B as a function of trial length, biomarker measurement error, carryover half-life, and between-participant heterogeneity. The full source for the exposition that motivated this paper is preserved in `notes/source-revision-latent-class.md`.

2 Introduction

Section stub. Frame the question in terms of the dual-tradition gap identified in the source revision response: regression-with- random-effects (biostatistical default) versus latent-variable construction (psychometric and econometric default). State the contribution: first principled application of latent-variable machinery to the biomarker-treatment interaction problem under the N-of-1 design family.

3 Statistical formulation

Section stub. Develop:

1. A two-class latent-class model with class membership predicted by the biomarker.
2. A finite-mixture-of-regressions model with biomarker-dependent mixing weights.
3. Connections to the random-coefficient growth-mixture-model tradition.
4. Identifiability conditions specific to the N-of-1 setting (within- participant treatment switches as a key source of identification).

4 Estimation

Section stub. EM algorithms, MCMC samplers, and the comparative behaviour of point versus posterior estimates. Software: `mclust`, `flexmix`, `OpenMx`, `Stan` candidate implementations.

5 Simulation study

Section stub. Re-use the trial-design grid and biomarker parameter space from the carryover-sensitivity factorial. Add a dimension over latent-class separability. Outcomes: power for the class-by-treatment

interaction, classification accuracy, recovery of true class means.

6 Application

Section stub. Re-analyse the prazosin-PTSD data with the latent- class formulation; compare to the Architecture A and B fits in the companion papers.

7 Discussion

Section stub. When does the latent-class construction add value over Architecture A or B? When does the data not support its identifiability?

8 Acknowledgements

9 References